

Supplementary Materials for  
*“Intrinsic Disorder Predisposition of the Lujo  
virus glycoprotein: A geometric algebra framework  
for viral protein characterization”*

## Supplementary Tables

**Table S1. Descriptive statistics for all 18 protein groups analyzed in this study.**

Internal cohesion (mean intra-group wedge product  $\pm$  SD), mean standard deviation across the 16 PIM components, adaptive similarity threshold (5th percentile of intra-group wedge products; for  $n = 1$  groups, threshold = 0.99), and number of sequences processed per group.

Table 1: Descriptive statistics for all 18 protein groups.

| Group            | Metric Similarity | Charge Angle (°) | Clifford Distance | Total |
|------------------|-------------------|------------------|-------------------|-------|
| CPP              | 0.95              | 29.44            | 0.20              |       |
| NON_CPP          | 0.89              | 27.51            | 0.05              |       |
| UNFOLDED         | 0.96              | 11.98            | 0.07              |       |
| JUNV             | 0.97              | 44.77            | 0.04              |       |
| LASV             | 0.96              | 4.25             | 0.05              |       |
| PARTIALLY_FOLDED | 0.99              | 21.25            | 0.03              |       |
| LCMV             | 0.97              | 2.50             | 0.05              |       |
| SIGNALS          | 0.98              | 5.20             | 0.06              |       |
| VIRUS_UNREVIEWED | 0.99              | 6.52             | 0.02              | 1,0   |
| UNREVIEWED_ALL   | 0.99              | 7.09             | 0.04              | 149   |
| VIRUS_REVIEWED   | 0.99              | 3.39             | 0.03              | 1     |
| UNREVIEWED_HUMAN | 1.00              | 8.71             | 0.03              | 1     |
| REVIEWED_HUMAN   | 0.99              | 5.86             | 0.03              | 2     |
| MEMBRANE         | 1.00              | 7.81             | 0.02              |       |
| REVIEWED_ALL     | 1.00              | 4.21             | 0.02              | 5     |
| DISEASE          | 1.00              | 8.25             | 0.03              |       |
| MACV             | 0.99              | 5.72             | 0.02              |       |

**Table S2. Ranked comparison of all 17 protein groups against the LUJV glycoprotein centroid.**

Ordered from highest to lowest wedge similarity. Values represent wedge similarity (mean  $\pm$  SD from bootstrap resampling with 100 iterations), wedge orientation (+1 = same orientation, -1 = opposite orientation), and hydrophobic rotation angle (degrees).

Table 2: Ranked comparison against LUJV glycoprotein centroid.

| Compared Group   | Wedge Similarity | Wedge Orientation | Hydrophobic Angle (°) |
|------------------|------------------|-------------------|-----------------------|
| CPP              | 0.12             | -1                | 26.45                 |
| NON_CPP          | 0.07             | -1                | 28.51                 |
| UNFOLDED         | 0.06             | -1                | 12.92                 |
| JUNV             | 0.04             | 1                 | 2.68                  |
| LASV             | 0.09             | -1                | 15.20                 |
| LCMV             | 0.06             | -1                | 13.79                 |
| PARTIALLY_FOLDED | 0.04             | 1                 | 6.70                  |
| UNREVIEWED_ALL   | 0.02             | 1                 | 1.98                  |
| SIGNALS          | 0.05             | 1                 | 8.99                  |
| VIRUS_REVIEWED   | 0.03             | 1                 | 5.91                  |
| VIRUS_UNREVIEWED | 0.02             | 1                 | 5.80                  |
| REVIEWED_ALL     | 0.03             | 1                 | 2.01                  |
| DISEASE          | 0.02             | 1                 | 3.24                  |
| MEMBRANE         | 0.02             | 1                 | 0.90                  |
| UNREVIEWED_HUMAN | 0.02             | 1                 | 3.81                  |
| REVIEWED_HUMAN   | 0.03             | 1                 | 4.25                  |
| MACV             | 0.04             | -1                | 5.62                  |

**Table S3. Pairwise wedge similarity matrix between centroids of all 18 protein groups.**

Values range from 0 to 1, where higher values indicate greater functional similarity between polarity profiles. The matrix is symmetric about the diagonal (differences  $\leq 0.001$ ), confirming computational consistency. Abbreviations: PF = PARTIALLY\_FOLDED; RH = REVIEWED\_HUMAN; UH = UNREVIEWED\_HUMAN; VR = VIRUS\_REVIEWED; VU = VIRUS\_UNREVIEWED; RA = REVIEWED\_ALL; UA = UNREVIEWED\_ALL.

Table 3: Pairwise wedge similarity matrix between centroids of all 18 protein groups.

| Group    | LUJV | LASV | JUNV | MACV | LCMV | PF   | CPP  | NON_CPP | UNFOLDED |
|----------|------|------|------|------|------|------|------|---------|----------|
| LUJV     | 0.00 | 0.09 | 0.04 | 0.03 | 0.06 | 0.04 | 0.20 | 0.14    | 0.06     |
| LASV     | 0.09 | 0.00 | 0.11 | 0.05 | 0.03 | 0.05 | 0.19 | 0.08    | 0.03     |
| JUNV     | 0.04 | 0.11 | 0.00 | 0.06 | 0.08 | 0.07 | 0.23 | 0.16    | 0.09     |
| MACV     | 0.03 | 0.05 | 0.06 | 0.00 | 0.03 | 0.02 | 0.18 | 0.10    | 0.03     |
| LCMV     | 0.06 | 0.03 | 0.08 | 0.03 | 0.00 | 0.02 | 0.19 | 0.09    | 0.01     |
| PF       | 0.04 | 0.05 | 0.07 | 0.02 | 0.02 | 0.00 | 0.17 | 0.09    | 0.02     |
| CPP      | 0.20 | 0.19 | 0.23 | 0.18 | 0.19 | 0.17 | 0.00 | 0.13    | 0.15     |
| NON_CPP  | 0.14 | 0.08 | 0.16 | 0.10 | 0.09 | 0.09 | 0.13 | 0.00    | 0.07     |
| UNFOLDED | 0.06 | 0.03 | 0.09 | 0.03 | 0.01 | 0.02 | 0.15 | 0.07    | 0.00     |
| RH       | 0.02 | 0.07 | 0.04 | 0.02 | 0.04 | 0.03 | 0.19 | 0.12    | 0.04     |
| UH       | 0.02 | 0.08 | 0.03 | 0.02 | 0.05 | 0.03 | 0.20 | 0.13    | 0.05     |
| SIGNALS  | 0.05 | 0.06 | 0.06 | 0.03 | 0.03 | 0.02 | 0.18 | 0.10    | 0.03     |
| MEMBRANE | 0.01 | 0.10 | 0.02 | 0.04 | 0.07 | 0.05 | 0.20 | 0.14    | 0.07     |
| DISEASE  | 0.02 | 0.08 | 0.03 | 0.03 | 0.05 | 0.04 | 0.20 | 0.13    | 0.05     |
| VR       | 0.03 | 0.07 | 0.04 | 0.02 | 0.04 | 0.02 | 0.20 | 0.12    | 0.04     |
| VU       | 0.03 | 0.06 | 0.04 | 0.02 | 0.04 | 0.03 | 0.20 | 0.12    | 0.04     |
| RA       | 0.02 | 0.11 | 0.02 | 0.05 | 0.07 | 0.06 | 0.20 | 0.15    | 0.07     |
| UA       | 0.03 | 0.11 | 0.02 | 0.05 | 0.07 | 0.06 | 0.20 | 0.14    | 0.07     |

**Table S4. Top 20 individual human proteins most similar to the LUJV glycoprotein centroid.**

Ranked by wedge similarity. Columns: Rank (order of similarity), Group (protein set of origin), Protein ID (UniProt identifier), Wedge Similarity (functional similarity score; 1 = identical, 0 = orthogonal), Orientation (+1 = same directional organization, -1 = opposite), and Hydro Angle ( $^{\circ}$ ) (rotation angle in the hydrophobic plane; smaller angles indicate better alignment of hydrophobic interaction patterns).

Table 4: Top 20 individual human proteins most similar to LUJV.

| Rank            | Group            | Protein ID         | Wedge Similarity | Orie |
|-----------------|------------------|--------------------|------------------|------|
| Hydro Angle (°) |                  |                    |                  |      |
| 1               | PARTIALLY_FOLDED | sp—PO9999—         | 0.551            |      |
| 90.0            |                  |                    |                  |      |
| 2               | PARTIALLY_FOLDED | sp—PO9999—         | 0.526            |      |
| 90.0            |                  |                    |                  |      |
| 3               | PARTIALLY_FOLDED | sp—PO9999—         | 0.509            |      |
| 40.08           |                  |                    |                  |      |
| 4               | PARTIALLY_FOLDED | sp—PO9999—         | 0.420            |      |
| 40.08           |                  |                    |                  |      |
| 5               | PARTIALLY_FOLDED | sp—PO9999—         | 0.400            |      |
| 49.92           |                  |                    |                  |      |
| 6               | PARTIALLY_FOLDED | sp—PO9999—         | 0.398            |      |
| 40.08           |                  |                    |                  |      |
| 7               | PARTIALLY_FOLDED | sp—PO9999—         | 0.396            |      |
| 40.08           |                  |                    |                  |      |
| 8               | PARTIALLY_FOLDED | sp—PO9999—         | 0.393            |      |
| 49.92           |                  |                    |                  |      |
| 9               | REVIEWED_ALL     | sp—P35904— (CDHR5) | 0.389            |      |
| 40.08           |                  |                    |                  |      |
| 10              | CPP              | —                  | 0.381            |      |
| 40.08           |                  |                    |                  |      |
| 11              | REVIEWED_ALL     | sp—P0DJB0— (NPAS3) | 0.375            |      |
| 40.08           |                  |                    |                  |      |
| 12              | CPP              | —                  | 0.374            |      |
| 49.92           |                  |                    |                  |      |
| 13              | SIGNALS          | sp—Q156A1—         | 0.373            |      |
| 49.92           |                  |                    |                  |      |
| 14              | PARTIALLY_FOLDED | sp—PO9999—         | 0.372            |      |
| 49.92           |                  |                    |                  |      |
| 15              | PARTIALLY_FOLDED | sp—PO9999—         | 0.372            |      |
| 49.92           |                  |                    |                  |      |
| 16              | CPP              | —                  | 0.371            |      |
| 49.92           |                  |                    |                  |      |
| 17              | CPP              | —                  | 0.371            |      |
| 49.92           |                  |                    |                  |      |
| 18              | CPP              | —                  | 0.371            |      |
| 49.92           |                  |                    |                  |      |
| 19              | CPP              | —                  | 0.371            |      |
| 49.92           |                  |                    |                  |      |
| 20              | CPP              | —                  | 0.371            |      |
| 49.92           |                  |                    |                  |      |

## Supplementary Methods

### Detailed SGP Algorithm Description

The SGP (Specular Grassmann-PIM) algorithm processes protein sequences through the following steps:

### Step 1 — Polarity classification

Each amino acid is mapped to one of four polarity/charge categories according to the PIM 3.0v scheme:

- Positively charged:  $P^+ = \{H, K, R\}$
- Negatively charged:  $P^- = \{D, E\}$
- Neutral polar:  $N = \{C, G, N, Q, S, T, Y\}$
- Non-polar (hydrophobic):  $NP = \{A, F, I, L, M, P, V, W\}$

### Step 2 — Construction of the polarity string

The polarity symbol of each residue in a sequence is used as a replacement. Example sequence (30 residues):

[A, I, M, T, A, A, I, N, C, S, E, E, T, C, S, A, I, M, T, A, A, I, N, C, S, E, E, T, C, S]

Polarity string:

[NP, NP, NP, N, NP, NP, NP, N, N, N,  $P^+$ ,  $P^+$ , N, N, N, NP, NP, NP, N, NP, NP, NP, N, N, N,  $P^+$ ,  $P^+$ , N, N, N].

### Step 3 — Counting adjacent transitions

Every consecutive pair of residues contributes to a  $4 \times 4$  transition matrix (rows = “from”, columns = “to”). For the example, total transitions = 29. The resulting transition frequency matrix is shown in **Table S5**.

**Table S5. Polarity transition frequency matrix for the 30-residue example peptide.**

Example sequence: AIMTAAINCSEETCSAIMTAAINCSEETCS.

Table 5: Polarity transition frequency matrix for the example peptide.

| From \ To | $P^+$ | $P^-$ | N | NP |
|-----------|-------|-------|---|----|
| $P^+$     | 0     | 0     | 0 | 0  |
| $P^-$     | 0     | 2     | 2 | 0  |
| N         | 0     | 1     | 8 | 2  |
| NP        | 0     | 0     | 4 | 10 |

### Step 4 — Normalisation and vectorisation

Each count is divided by the total number of transitions (29), resulting in a 16-component probability vector (the PIM profile). The 16 components are obtained by unrolling the matrix row by row.

### Step 5 — Grassmann wedge product (functional similarity)

The wedge product of two vectors  $v$  and  $w$  is the cosine of the angle between them:

$$v \wedge w = \frac{v \cdot w}{\|v\| \|w\|}$$

A value of 1 means identical functional profiles and a value of 0 orthogonality.

### Step 6 — Rotor angles (Clifford rotor analysis)

The vectors are projected onto six biologically relevant planes (e.g., the hydrophobic plane defined by components N→N and NP→NP). The rotation angle  $\theta$  required to align the two projections is computed as:

$$\theta = \arccos \left( \frac{p_1 \cdot p_2}{\|p_1\| \|p_2\|} \right) \text{ (in degrees)}$$

### Step 7 — Specular reflection detection

A “reflected” version of vector  $v$  is created by swapping the roles of  $P^+$  and  $P^-$  (indices  $0 \leftrightarrow 5$ ,  $1 \leftrightarrow 4$ ,  $2 \leftrightarrow 6$ ,  $3 \leftrightarrow 7$ , etc.) and applying the real geometric algebra reflection formula:

$$v' = v - 2(v \cdot n)n$$

If the wedge product between this reflected vector and a target vector  $w$  exceeds 0.95, the two proteins are considered specular reflections of each other, indicating a charge-swapping functional relationship.

### Step 8 — Clifford signature

For each group centroid, a compact 6-number fingerprint is calculated, comprising:

- Norm (vector magnitude)
- Auto-reflection (similarity with its own charge-swapped version)
- Hydrophobic projection (norm of components 10 and 15)
- Charge projection (norm of components 0 and 5)
- Auto-rotation (wedge product with a 4-position rolled copy)
- Total bivector (average of selected bivector components)

### Step 9 — Adaptive threshold and anomaly detection

The similarity threshold for each group is calculated automatically as the 5th percentile of intra-group wedge products. A query protein is classified as an anomaly if its best match has a wedge product below the adaptive threshold of its closest group and a Mahalanobis distance-based probability of belonging lower than 0.5.

## Supplementary Figure

### **Figure S1. Additional histograms of wedge product similarity distributions**

Additional histograms of wedge product similarity distributions between LUJV and specific protein subgroups are available from the corresponding author upon request.

**Supplementary Materials availability:** All supplementary tables, methods, and figures are available in this document. Additional data and custom programs are available from the corresponding author upon reasonable request.